SV16 to-do list - rebuild step 1

Use the step-1 copy SBSV-16_rebuild_step1_values.kicad_pcb as the starting file. Tick each item when done. Items marked (not bought) need purchase before the board can work.

A. Footprint changes in KiCad (Footprint Editor / Change Footprint)

[ ] C25, C30: change footprint from C_0805 (ceramic) to CP_Elec_4x5.4 (electrolytic), value 10 uF 35 V.

[ ] C32: change footprint from C_0805 to C_0603 (470 nF).

[ ] R9, R10, R11, R12 (100 ohm): change footprint from R_1206 to R_0805 (bought 100 ohm pack is 0805).

[ ] L1, L2 (10 uH CD54): change footprint from L_Sunlord_SWPA6045S to L_Sunlord_MWSA0518 (5.4x5.2 mm). Check pads against the bought part datasheet.

[ ] Y1 (oscillator, ECS-3225MV-250): change footprint from Oscillator_SMD_Abracon_ASE-4Pin_3.2x2.5mm to the ECS land pattern: 4 pads, 1.4 x 1.2 mm, centres at (+-1.10, +-0.80) mm. Dimensions read from the ECS datasheet figure labels: check against the ECS drawing before use. Pad 1 = tri-state, 2 = GND, 3 = OUT, 4 = VDD.

[ ] Y2 (12 MHz): change footprint from Crystal_HC49-4H (through-hole) to the SMD HC49/US footprint for the bought surface-mount crystal.

[ ] FB1 (600 ohm ferrite): change footprint from 0603 to 0805 (Inductor_SMD:L_0805_2012Metric). Part in cart: TDK MPZ2012S601ATD25, 0805, qty 2.

[ ] C36, C37 (22 pF): change footprint from C_0603 to C_0805 (YAGEO CC0805JRNPO9BN220, C0G 50 V 5%).

[ ] R5 and R25 (4.7k): change footprint from R_0603 to R_0805 (use the 4.7k 0805 pack you bought).

[ ] R6, R7, R8 (1k): change footprint from R_0603 to R_1206 (use the 1k 1206 pack you bought).

[ ] R47 (10k): change footprint from R_1206 to R_0603. Use the 5% 0603 part (YAGEO RC0603JR-0710KL), already in cart.

[ ] J7 (2x25 spare header): add Connector_PinHeader_2.54mm:PinHeader_2x25_P2.54mm_Vertical and mark it DNP (not fitted).

[ ] D6 (red LED, INITN): mark DNP. J2 (JTAG_ALT): mark DNP (spec: unpopulated).

[ ] J6 (motor header): shrouded, keyed 1x6 header. Footprint stays 1x6.

[ ] R21, R22, R23, R24 (0 ohm): keep the 0603 footprint. Use as solder-bridge links, populate one pair only (R21/R23 or R22/R24). No 0 ohm bought.

B. Schematic edits (you make these in the schematic, then update the board from it)

[ ] U6 COMP network: change R45 from 10k to 150k, and C35 from 100 nF to 100 pF. Calculated with the MP1584EN datasheet method: fsw 911 kHz (R40 = 100k), crossover about 91 kHz, output C21 = 100 uF, Vout 1.1 V. At 10k the crossover would be about 6 kHz, far below the expected range.

[ ] U6 COMP second capacitor (C6 on COMP): do NOT fit. The datasheet table lists no second COMP capacitor for low output voltages. Verify loop stability on the bench.

[ ] C23 (470 uF input bulk): change the voltage to 25 V (electrolytic, CP_Elec_8x10.5). 16 V on a 12 V rail is 75% of rated voltage, too close to the limit.

[ ] 5V_USB: connect this net to USB_VBUS in the schematic. The spec says the 5 V pins on J3 (17, 18) and J4 (20) are USB VBUS only. Right now 5V_USB has no source, so those pins are unpowered.

[ ] U5 (AP62300) FB: keep the divider, R46 = 33k from 3V3 to U5_FB and R47 = 10k from U5_FB to GND. Do NOT connect U5_FB to 3V3. Output = 0.763 x (1 + 33/10) = 3.28 V. The spec correction that said to tie FB to 3V3 is wrong for this part.

C. Values to set (check these in KiCad after the schematic is updated)

[ ] R44 = 1k (LED D7 current limit, 3V3 to D7). Keep.

[ ] R48 = 100k (EN top, VM_IN to U5_EN), R49 = 33k (EN bottom). Keep. At 12 V, EN = 2.98 V, under the 6 V EN pin limit. EN threshold 1.2 V typ (Diodes DS41958), so the buck starts at about 4.7 to 4.9 V VIN.

[ ] C34 = 100 nF (CH340G V3 decoupling). Keep.

[ ] R38 = 12.4k, R39 = 33k: U6 feedback. Output = 0.8 x (1 + 12.4/33) = 1.10 V. Keep.

[ ] Q2 = MMBT3904 and Q4 = MMBT3906 (SOT-23, pinout per spec). C36, C37 = 22 pF.

[ ] Resistors: R1, R13-R19, R26, R32, R47 = 10k 5%. R2-R5, R25 = 4.7k. R6-R8, R31, R33, R34 = 1k. R9-R12 = 100 ohm. R27-R30 = 470 ohm. R36, R37 = 5.1k. R40, R41, R43 = 100k. R46 = 33k.

[ ] Check the file opens with all 141 footprints. Confirm the component positions are unchanged.

D. Circuit verification (from sv16_board.kicad_sch, checked against the spec)

[ ] PASS: FPGA power pins match the spec. 9 VCCIO pins on 3V3, 4 VCCAUX pins on 2V5, 6 VCC pins on 1V1, 14 GND.

[ ] PASS: decoupling per rail. 3V3: C12-C20 plus C25, C26-C30, C40, C41. 2V5: C8-C11, C22, C42. 1V1: C2-C7, C21.

[ ] PASS: AP62300 (U5): EN divider 100k/33k, BST cap C38, SW to L1 (10 uH) to 3V3, VIN bulk C39 plus C23.

[ ] PASS: MP1584 (U6): FB divider 12.4k/33k gives 1.10 V, FREQ resistor R40 = 100k gives 911 kHz, EN delay R41 + C31, BST cap C33, catch diode D11 (SS34, cathode on SW), L2 10 uH to 1V1.

[ ] PASS: LP5907 (U7): EN delay R43 100k + C32 470 nF. The 2V5 switch matches the spec correction 4 (Q3 and R42 removed).

[ ] PASS: USB-C (J8): CC1/CC2 each have a 5.1k pull-down (R36, R37). VBUS to VM_IN_RAW through D9 and FB1. D+/D- through the USBLC6 array (U8) to the CH340G.

[ ] PASS: CH340G (U4, keep this part): VCC on 3V3, 12 MHz crystal Y2 with 22 pF C36/C37, V3 decoupling C34, USB D+/D- on pins 5/6.

[ ] PASS: Y1 pinout matches the ECS-3225MV (1 enable tied to 3V3, 2 GND, 3 clk_25m, 4 VDD on 3V3).

[ ] PASS: flash U2/U3 WP and HOLD pins have pull-ups to 3V3 (R14, R15, R17, R18).

[ ] ACTION: 5V_USB is not tied to USB_VBUS. See B.

[ ] ACTION: R45 = 10k and C35 = 100 nF in the schematic. See B.

[ ] ACTION: U5 FB must stay on the divider (not 3V3). See B. This corrects the spec correction 1 that I passed on earlier.

[ ] ACTION: C23 voltage is 16 V. See B.

[ ] VERIFY: CH340G pin 15 (R232) and pins 9-12, 14 (modem control, unused). Check the CH340G datasheet for the required state of pin 15.

[ ] VERIFY: LP5907 pin 4 is unconnected in the schematic. Check the LP5907 datasheet (BYP, optional noise cap).

[ ] VERIFY: J1 (JTAG) pin 5 is unconnected in the schematic. Check it against the JTAG connector standard for your programmer.

[ ] VERIFY: the FPGA has 48 unlabelled pins. The spec lists 46 unassigned I/O pins. Check the 2 extra are NC or spare, not power.

[ ] RISK: the spec says the AP62300 and LP5907 want ceramic output caps (spec correction 6). You chose electrolytic for C39-C42. Check the LP5907 stability on the bench. Consider ceramics if it is unstable.

E. Purchases still needed (not in any cart screenshot)

[ ] U1: FPGA LFE5U-12F-6TG144C (not bought).

[ ] U3: W25Q64JVSSIQ, 1 in cart. With the 1 you have, that makes 2 flash chips (U2, U3). Covered.

[ ] Y1: ECS-3225MV-250, active 25 MHz XO 3.2x2.5 4-pad. The 3 bought 2-pin 25 MHz crystals cannot be used here.

[ ] Y2: 12 MHz SMD crystal HC49/U (surface mount, 2-pin). In cart (qty 2).

[ ] D8/D9/D11: 1 more SS34 (you have 2, need 3). SS34 is in cart (qty 1).

[ ] 10k 0603 5% resistors: YAGEO RC0603JR-0710KL, 12 pieces (R1, R13-R19, R26, R32, R47). In cart (qty 110).

[ ] 22 pF 0805 C0G caps: YAGEO CC0805JRNPO9BN220, 2 pieces (C36, C37). In cart (qty 4).

[ ] FB1: TDK MPZ2012S601ATD25, 0805, 600 ohm. In cart (qty 2).

[ ] R44 (1k 0603): check stock. Buy if you have none.

[ ] R45 (150k 0603): buy. Not in any cart.

[ ] C35 (100 pF 0603 C0G): check stock. Buy if you have none.

[ ] 4.7k (R5, R25) and 1k (R6-R8): use the 0805 and 1206 packs already bought. No purchase needed.

[ ] Headers: standard 2.54 mm straight male pin headers (1x10, 2x10, 1x6, 1x4, 1x2). J6 is the exception: shrouded, keyed 1x6.

[ ] Connectors and switches (barrel jack J9, SW1/SW2, JP1 shunt): you already have these. Confirm the parts and footprints match the board.

F. Handoff notes

[ ] Starting file: SBSV-16_rebuild_step1_values.kicad_pcb (in this folder). It has 141 footprints, the same positions and a 4-layer stackup. 37 values changed. Do not move components. Do not change pad numbers or net names.

[ ] Pin 1 of U6 = SW, 2 = EN, 3 = COMP, 4 = FB, 5 = GND, 6 = FREQ, 7 = VIN, 8 = BST. Confirmed against the schematic.

[ ] Net names: the file keeps USB_DP/USB_DN and 5V_USB. The spec names are USB_D+/USB_D- and 5V. Record this in REPORT.md.

[ ] Logo footprint G*** is removed in the step-1 file. Leave it removed and record it in REPORT.md. J3 must stay at its original position (194.65, 120.24 mm).

[ ] Design rules (spec section 7): signal track 0.25 mm; 3V3, 1V1, 2V5 power tracks 0.8 mm; VM_IN 1.5 mm; switching nodes 0.8 mm and short. Clearance 0.2 mm minimum.

H. Still open (summary, tick when resolved)

[ ] Schematic: change R45 from 10k to 150k (COMP resistor).

[ ] Schematic: change C35 from 100 nF to 100 pF (COMP capacitor).

[ ] Schematic: change C23 to 25 V.

[ ] Schematic: connect 5V_USB to USB_VBUS (5 V header pins are VBUS only).

[ ] Footprint: confirm the Y1 ECS land pattern (1.4 x 1.2 mm, centres +-1.10 / +-0.80 mm) against the ECS drawing.

[ ] Datasheet: CH340G pin 15 (R232) and the unused modem pins 9-12, 14.

[ ] Datasheet: LP5907 pin 4 (BYP) is unconnected in the schematic. Confirm it can stay open.

[ ] Datasheet: JTAG J1 pin 5 is unconnected. Confirm against your programmer standard.

[ ] FPGA: 48 unlabelled pins vs 46 unassigned I/O in the spec. Confirm the 2 extra are NC or spare.

[ ] Risk: electrolytic output caps (C39-C42) vs ceramics in the spec. Check LP5907 and AP62300 stability on the bench.

[ ] Regenerate the netted board from the updated schematic and BOM (not done yet).

[ ] Electrical rule check in KiCad (not run here; no KiCad CLI in this environment).

[ ] Bench test: power-up, rail voltages (3V3 3.28 V, 1V1 1.10 V, 2V5), clock on clk_25m, USB enumeration, flash read.

G. Routing and checks (later step, not part of this netted board)

[ ] Strip the old routing first: the step-1 file still has 1,783 tracks and 179 vias. Delete all tracks and vias. Keep every footprint in place. Do not merge or copy old routing.

[ ] Re-route from the updated placement, then run DRC: 0 unconnected, 0 shorts, 0 clearance, 0 edge or courtyard errors. Zones filled.

[ ] Only after clean DRC: generate Gerber, drill, BOM and position files.